

Resilient decentralized communication for unstructured peer-to-peer networks

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Abstract

Recent crises, natural disasters, and large-scale protests have shown that people heavily rely on online social networks and instant messaging platforms to communicate. During crises, communication networks can be strained, partially or wholly destroyed, censored, or otherwise monitored, leaving reliant communities in need of alternatives. An alternative to these centralized, concentrated, and consolidated networks is decentralized peer-to-peer overlay networks. In this paper, we provide empirical insights to parametrize simulations of local area networks at scale based on Bluetooth. We then define resilience based on sparsity to capture the performance of the overlay. We show design considerations for the overlay to outperform state-of-the-art overlay designs based on resilience and communication latency.

1 Introduction

The decentralized administration of the internet is one of the basic enablers of the rapid growth of the Internet as a whole, thanks to the autonomous nature of decision making across different network systems. The core functionality of the Internet is based on a general-purpose solution, which dictates the establishment, maintenance, and termination of connections between the different Internet autonomous system (ASs) using special border routers between them. The ISPs have a central critical role in the functionality of the Internet, as they provide connection to the Internet backbone to customer end-users. The power dynamic of this system inherently gives control over consumption patterns to the ISPs; consolidation of power at the hands of a few ISPs gives them the unique ability to control the flow of information, throttling, and monitoring, in addition to enabling oppressive practices such as censorship and surveillance end users. Additionally, an imminent danger to the freedom of speech and freedom of access to information is the increasing trend of media corporations such as Meta and X to employ both explicit and implicit content moderation practices. Together, the aforementioned practices have in recent years lead to a growing interest in decentralized peer-to-peer social networks and messaging platforms. Most notably, the rise of fully-decentralized networks like Mastodon, user-supported decentralized networks such as Bluesky, and end-to-end encrypted decentralized messaging platforms like Briar, Signal, and until recently Firechat, have proven that there is a growing shift in user interest patterns away from traditional Centralized Social Networks (CSNs) towards decentralized social networks of diverse approaches. In this research paper, we focus on measuring the performance, resilience, and sparsity metrics of unstructured peer-to-peer networks. We decided to focus on unstructured peer-to-peer networks as they exhibit a higher

level of information decentralization, are more resilient to topology changes caused by a high node churn, and are more suitable for the dynamic communication natures. We present both empirical performance characteristics as well as simulated resilience metrics, using a physical testbed of devices communicating over BLE, as well as simulator designed for opportunistic network environments, to evaluate metrics based on the empirical evidence we collected.

2 Background

There is a growing architectural gap between the internet's general-purpose design and the individual implementation of ISPs in their ASes, and the increasing demand for specific functionalities (e.g. Multicast) makes it necessary to implement change when things break [2]. This is because the Internet was made with the concept of one-size-fits-all, allowing different Internet providers to implement different rules, policies, and techniques to facilitate their networks. The general-purpose design of the Internet also introduces the immutability of paths between pairs of nodes - once a path between two nodes exists, and the nodes can use it to reach each other, it is a complex process to get the nodes to communicate over new addresses, which happens when those nodes move. In the modern age nodes oftentimes exhibit movement patterns that may render their previous Internet connection point unusable, thereby forcing the nodes to search for a new attachment point - oftentimes switching from WiFi to cellular, between WiFi networks, or cellular to WiFi. In disaster zones, for example after earthquakes, fires, and other natural disasters, existing networking infrastructure is rendered useless due to destruction. In less-developed areas of the world, such infrastructure might not be available at all: the Lapponia region near the Arctic Circle, homeland of the Sámi people, is characterized as a sparsely-populated region full of icy fjords, deep valleys, glaciers and mountains. The inhabitants of this region have traditionally been reindeer-herders, and often migrate with the reindeer herds along the yearly cycle. Due to the harsh terrain, sparse level of population, and partial autonomy of the Sámi council, the Sámi people lack reliable cellular and internet infrastructure. To provide the Sámi people with modern communication options, the traditional packet-switching paradigm of communication as conceptualized by the Internet is not optimal. Instead, Lindgren et al. propose the Store-Carry-Forward paradigm using Delay-Tolerant Networking (DTN) [3], where hosts store messages until an opportunity arises to forward them, carry them while moving until they reach a rendezvous with another node or a network attachment point, and then forward them upon contact. This paradigm enables the opportunistic usage of technologies such as Bluetooth to forward

messages, which is ubiquitous to virtually all Smartphones and computers today. Insofar as the Internet is not available, DTNs using Store-Carry-Forward networking can be used as a fallback, being supplemented by traditional Internet infrastructure as it becomes available. Decentralized social networks can exist either as unstructured or structured peer-to-peer networks, as a democratic ability to run the entire software stack of a server individually (cf. Mastodon), or as a collection of transparent community-operated competing components belonging to the same social network, like in Bluesky. Structured peer-to-peer networks such as Chord, Kademlia, and the InterPlanetary File System (IPFS) use Distributed Hash Tables (DHTs) to store and retrieve content in a decentralized manner. On the other hand, unstructured peer-to-peer networks are more dynamic and do not use DHT to spread centralized information. Instead, unstructured peer-to-peer networks rely on flooding and gossip for content dissemination, like in Gnutella and BitTorrent. This concept has been the guiding idea of this research paper, with the intent to investigate the reliability and performance of unstructured peer-to-peer networks and grassroots social networks built using flooding and gossip for neighbor discovery and content dissemination.

3 Related Work

Decentralised administration is one of the basic enablers of the growth of the Internet as a whole, thanks to the autonomous organization of different Internet Service Providers. Different countries and geographical regions around the world have different characteristics of centralization around their service providers, which affects the Internet usage patterns of their residents, their resilience, and actually how well they can mitigate outages, by using a mixture of both endemic as well as external providers. In [5] Habib et al. study patterns of web service hosting in different countries, focusing on websites, CA authorities, and more. Habib et al. demonstrate how countries with higher centralization around a smaller number of hosting providers are more susceptible to outages, and propose a metric called the Centralization Score S , which measures the centralization of website hosting around a small number of providers, and derive it from the Herfindahl-Hirschmann Index (HHI), which is used to measure market competition in US antitrust laws. Habib et al. showed how Thailand, that has over 60% of all websites hosted by its most popular provider, starkly contrasts with Iran, which has a more distributed hosting landscape with its top provider account for only 14% of all websites, while 90% of all websites are distributed across 80 different providers. To use the different sparsity metrics we will introduce, we defined node degree, or the number of connections a node has to other nodes in the topology, as the sparsity measurement: the more similar the node degree ranks are to each other, the more distributed and less sparse a topology is, and vice-versa: more sparse node degrees make the topology more centralized, as a relatively small number of hosts are connected to a large amount of nodes, and most nodes are connected to a small amount of nodes. The Centralization Score S is defined as follows:

$$S = \sum_{i=0}^n \left(\frac{a_i}{C} \right)^2 - \frac{1}{C}$$

where C is the total number of websites considered, a_i is the number of websites hosted by provider i , and n is the total number of providers. In this paper, we define C as the total number of links between nodes, a_i is the node degree of node i , and n is the total number of nodes. Hurley and Rickard compare different measures of sparsity[4], and rank different known sparsity measures according to six desirable criteria that a sparsity measure should satisfy, among them the Robin Hood axiom ("stealing from the rich and giving to the poor": distributing websites over multiple providers instead of a single provider reduces sparsity), the Scaling axiom (sparsity is scale-invariant), and the Rising Tide axiom (sparsity is decreased by adding a constant to all coefficients). Hurley et al. prove that amongst the most common methods to measure sparsity, the Gini coefficient outperformed all other measurements, and satisfies all six criteria. The Gini coefficient, a metric commonly associated with the Human Development Index to measure income inequality around the world, is calculated as follows:

$$G = \frac{\sum_{i=0}^n \sum_{j=0}^n |a_i - a_j|}{2n \sum_{i=0}^n a_i}$$

Where a_i is the node degree of node i , and n is the total number of nodes. Hurley and Rickard's work allows us to effectively use tried-and-tested economical metrics to measure the sparsity of our experiment topologies. In this paper, we use the Centralisation Score S to compare different topologies and rank them according to their centralization and sparsity metrics.

4 Setup

4.1 Testbed

To measure the performance characteristics, we decided to measure the Received Signal Strength Indicator (RSSI) of BLE advertisements as received by peer nodes in the testbed, in increasing distances, to measure the signal coverage, and how it propagates in free space. Initially contemplating WiFi as well, we decided to focus on Bluetooth as it is more power efficient, and requires no network infrastructure. Furthermore, Bluetooth Low Energy (BLE) was picked over classic Basic Rate/Enhanced Data Rate (BR/EDR), as it is even more efficient for (small) message exchanges. BLE advertisements form the bottleneck of BLE communications, as they are used to discover neighbours and establish connections between nodes in an unstructured peer-to-peer network based on BLE. To discover peers, nodes have to actively scan for BLE advertisements from other nodes, and not advertise at the same time, due to the half-duplex nature of BLE.

Our testbed consists of a set of Raspberry Pi 3B+ devices, using the built-in Bluetooth 4.2 as well as LE supporting chip Cypress CYW43455, and external power supplies of 5v/3.4A to ensure stable operation and mobility. The Raspberry Pis were flashed with a lightweight 64-bit Linux port (Raspberry Pi OS Lite based on Debian Trixie), upon which the basic Bluetooth stack was installed using BlueZ 5.66. The intention in this part of the paper, was to provide empirical evidence of the signal performance characteristics of BLE, so that they may be used as parameters in the ONE simulation.

4.2 Simulations

To characterise the sparsity and resilience metrics and characterise the delivery statistics of Bluetooth-based communication, we used The ONE¹ simulator to simulate the testing environment. A large 480×480 L-shape building hall was generated in WKT format to be used by the simulator, split to a grid of 12 equal-sized cells of 120 × 120 meters each assigned to a cluster consisting of six hosts, for a total of 72 hosts. Each host was randomly placed within the cluster cell assigned to it. A simulation topology is generated by a random number generator that generates the node positions, whose seed is derived from the communication range and the message size, to ensure reproducibility of results. The communication range is a configurable parameter which limits the radius of communication around a node: for simplification, a connection can be established between two nodes if they are within direct line of sight of each other and in the communication radius. A connection does not churn during a simulation run: once a connection is established it is maintained until the end of the simulation run. Different topologies were generated by using the communication ranges of [1,5,10,20,30,40,50,60,70,80,90,100,110, 120] meters. A simulation run configuration consisted of a set of parameters, including the message generation rate, the maximum communication range of each node, the maximum number of parallel connections (max node degree) each node can establish and maintain, and the communication mode: either exclusively intra-cluster communication to simulate low-range cluster communication, or inter-cluster communication to simulate long-range communication. The same exact topology was repeated for communication ranges of [1,2,3,4,5,6,7,8,9,10,15,20] maximum node degrees, each for 25 runs, to assess how the maximum number of parallel connections of every node, or maximum node degree, affects the centralization and sparsity metrics of peer-to-peer networks operating on BLE. Between each run of the same configuration the topology does not change, but the order of the hosts picked for the message generation is shuffled, so that no bias is introduced by the ordering of the host IDs. In each run, every node remains static (does not move) and attempts to send the same amount of messages to all other nodes in the network, and an Epidemic Router was used to flood messages across the network, to simulate the scenario of message propagation in disaster zones, where it is critical to disseminate information as fast and as wide as possible, and no routes can be forecasted or assumed to exist. In an epidemic routing scheme every node sends all its messages to all of its neighbors, and each neighbor recursively repeats this iterative process until the final destination node for each message is reached. A simulator run ends when all generated messages are either delivered or discarded.

We wanted to evaluate the performance of the Bluetooth Low Energy running on 1M PHY mode², which means a symbol modulation rate of 1Mbps³ at ≤ 1 meter, assuming free space propagation. For the signal path loss model, we used the logarithmic path loss model[1], to interpolate the RSSI at continuous distances between source and destination node pairs based on the empirical evidence.

The path loss model is defined as follows:

$$RSSI_d(\text{dBm}) = A - 10 * \alpha * \log_{10}(d)$$

where $RSSI_d$ is the received signal strength indicator at distance d between source and destination nodes, A is the RSSI at the nearest lower or equal reference distance d_0 (as measured by the testbed), α is the path loss exponent (2 in free space), and d is the distance between the distance point to interpolate and the nearest empirical measurement. Using the RSSI measurements collected from the testbed, we derived the parameters for the path loss. The simulation was repeated for 25 runs per configuration, using an Ubuntu 22.04 with a Intel(R) Xeon(R) Gold 6326 CPU @ 2.90GHz with 64 cores, and 247 GB of RAM. To run the simulations we used parallelization by spawning 60 parallel jobs, each scheduled to a different CPU core.

5 Experiment

5.1 Empirical Testbed Experiment

In the empirical testbed measurement experiment, we positioned a pair of testbed nodes in an increasing distance from each other, on an outdoor olympic running track, starting from 0 meters (the nodes were located on top of each other) up to 120 meters, in an open field with minimal obstacles to signal propagation. At first, one node was powered on and fixed static to a location, and set as the advertiser node. The other node was placed in an appropriate distance, and a Python script was used to scan for BLE advertisements from the first node. The RSSI value, transmitting power, and the timestamp at time of reception were logged for each advertisement received. For every such configuration the logging node scanned for two minutes, paused for a short while to let the antenna cool down, and repeated the scanning for twenty times. As an advertising node, the tool *bluetoothctl* was used to register BLE advertisements, which advertised the node name, transmitting power, and most importantly set the advertisement interval. Due to technical hardware reasons, we are not able to control the transmission power of the BLE advertisements, and it was recorded to be the same across all measurements: 12 dBm. The wireless chip Cypress CYW43455, as supported by our testbed nodes Raspberry Pi 3B+, does not let us change the used transmission power; a point for future research could be to measure the maximum achieved distance with varying transmitting power levels using a different chip or a wireless dongle. Furthermore, we were not able to have fine-grain control over exactly when the advertisements were sent, because the BlueZ Bluetooth communication stack and the underlying hardware controller handle the advertisement scheduling internally to prevent antenna overheating. To best mitigate this, we set the advertisement interval window to the minimum possible values of [20 ms, 20 ms] to achieve the highest consistent frequency of advertisements sent. We were however able to prove that the advertisements were not actually being consistently sent with the configured interval window of 20ms using a Bluetooth sniffer on a mobile phone, which measured the actual advertisement intervals: these fluctuated around 20-100ms. Another point for future research would be to develop a Bluetooth driver that allows fine-grain control over the advertising scheduling; for that the python-dbus library could be

¹<http://akeranen.github.io/the-one/>

²<https://www.bluetooth.com/blog/exploring-bluetooth-5-how-fast-can-it-be/>

³https://dev.ti.com/tirex/explore/node?node=A__AQgtSgkM1Yu1Yimn5wFTOW__SIMPLELINK-ACADEMY-CC23XX_gsUPh5j__LATEST

used to directly interface with the underlying communication data bus, D-Bus.

5.2 Results

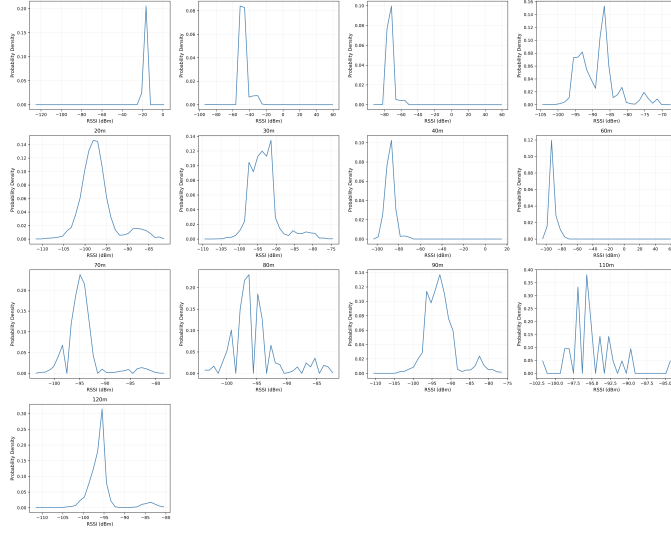


Figure 1: RSSI KDE plot at different distances

Figure 1 shows the Kernel Density Estimation (KDE) plots of the RSSI values recorded at different distances, using 30 bins for the RSSI values. We can see that most distances exhibit a normal distribution of values where most values are centered around the mean, with some outliers, while some distances exhibit a jittery behavior, attributed to passing temporary environmental interference. Generally, it is easy to see a correlation between an increase in the distance and a decrease in the RSSI value.

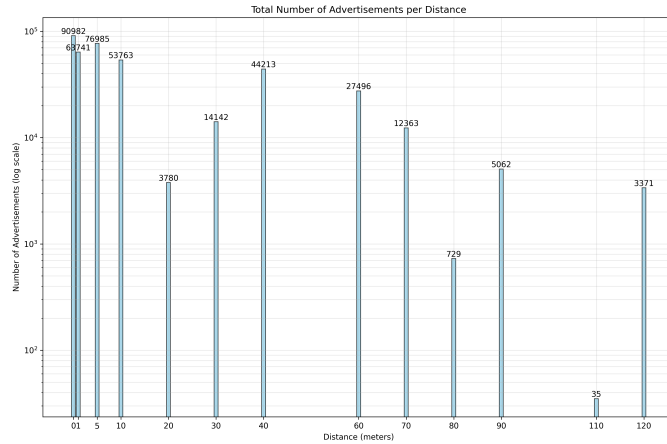


Figure 2: Number of advertisements received at different distances

Figure 2 shows the number of advertisements received at different distances. There is strong negative trend between the distance

and the number of advertisements received, as expected. We can also here observe a sudden unexpected drop in the number of advertisements received (for example, at 20 meters), which we again attribute to temporary environmental interference (e.g. runners at the track passing by the experiment obstructing the clear path, sometimes with phones or other metal objects in their pockets that reflect the signals and cause more wireless interference).

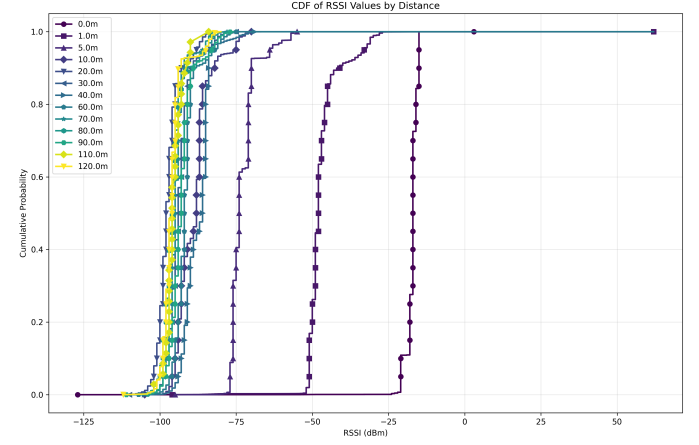


Figure 3: CDF of the RSSI vs. distance

Figure 3 shows the CDF of the RSSI values at different distances. On the X axis we can see the RSSI values in dBm, and on the Y axis the cumulative distribution function (CDF) of the RSSI values recorded at that distance.

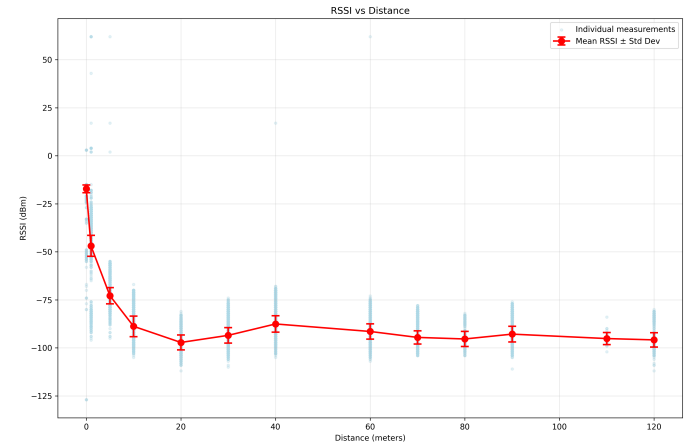


Figure 4: RSSI vs. distance

Figure 4 shows how the RSSI changes with distance. Each point represents the mean RSSI value recorded at the distance specified, with the bars indicating standard deviation. This plot clearly shows an expected power law decrease of the RSSI as the distance increases.

5.3 Simulations

After finishing the empirical testbed data collection experiment, we used the collected data to derive the signal parameters for the Bluetooth signal path loss model, that was used in the ONE simulator to simulate different experiment topologies. We used an Red-Black tree to store the distance value as keys, with the RSSI values as values, and interpolated for distance points between the keys.

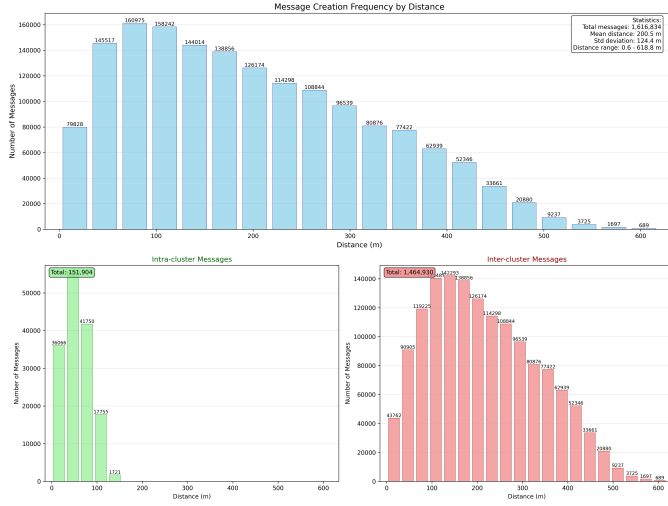


Figure 5: Message distance distribution

Figure 5 shows the distribution of the message distances that were generated. In every run, every host attempted to message every other host in the simulation, regardless of whether that host was directly reachable.

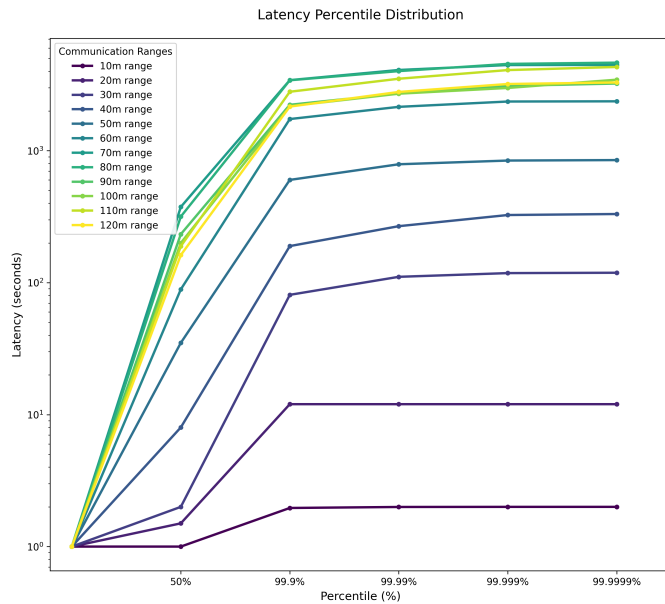


Figure 6: Latency percentiles

The figure 6 shows the latency measured in seconds, of delivering a message, as depending on the communication range. The different communication radii are shown as distinct curves on the plot, and shows as expected a trend towards a lower latency with a lower communication radius. Lower communication radii in turn create less messages, which exhibit less strain on the message buffers of the individual hosts.

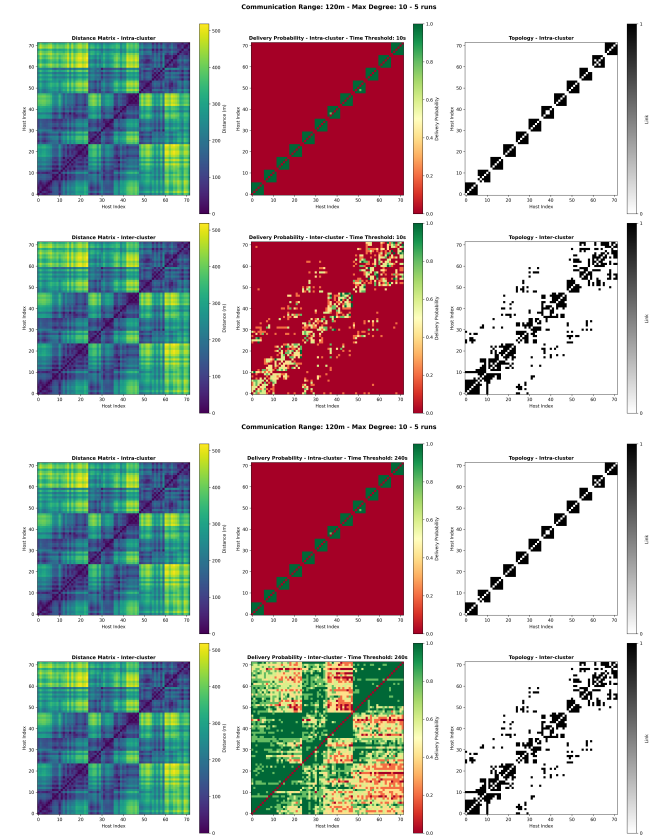


Figure 7: Topology overview - 120m range

The figure 7 shows two identical topologies, in which the communication range, or the communication radius of the network interface of every node, has a range of 120 meters. The difference between them is the time threshold used for the middle plot; in the first plot, the threshold of 10 seconds was used, and in the latter the threshold was set to 240 seconds. The center plot, the delivery probability plot, measures the probability of a message being delivered within the time threshold, as a heatmap between every pair of nodes in the network. The color indicates the delivery probability: the delivery probability within that threshold becomes lower the more red the color is, and is greener when the delivery probability improves. A delivery probability of 1.0 indicates all messages have been delivered within that threshold, and a delivery probability of 0.0 indicates that no message was delivered within the threshold. The right matrix is the topological matrix: in this matrix, a connection between two nodes is indicated by the solid black color. Finally, the left matrix shows the distance between every two nodes: a more

intense yellow indicates a higher distance, and a dark blue indicates less distance. It is easy to see that nodes within the same cluster are the closest to one another, and the clusters are the most visible in the intra-cluster topological matrix.

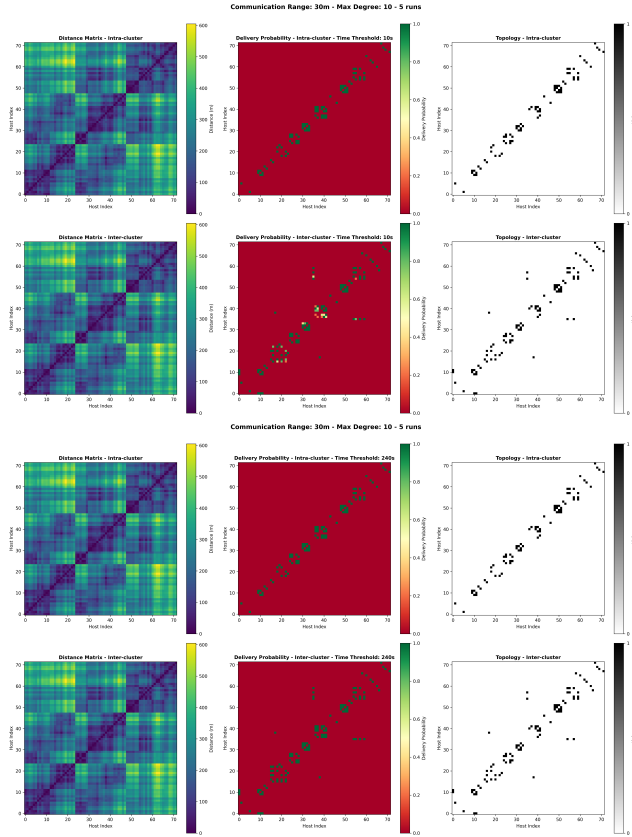


Figure 8: Topology overview - 30m range

Similarly the figure 8 shows the same plots, but exhibits a different topology, as the host positions were randomized due to having a different communication range. As mentioned in previous chapters, the random number generator seed is determined by the communication radius and the message size, which was kept static. In comparison, the 120m delivery probability plot at 7 exhibits a much higher delivery probability on average, as indicated by brighter green colors, in comparison to the 30m range. These matrices were generated for every different topology and maximum node degree.

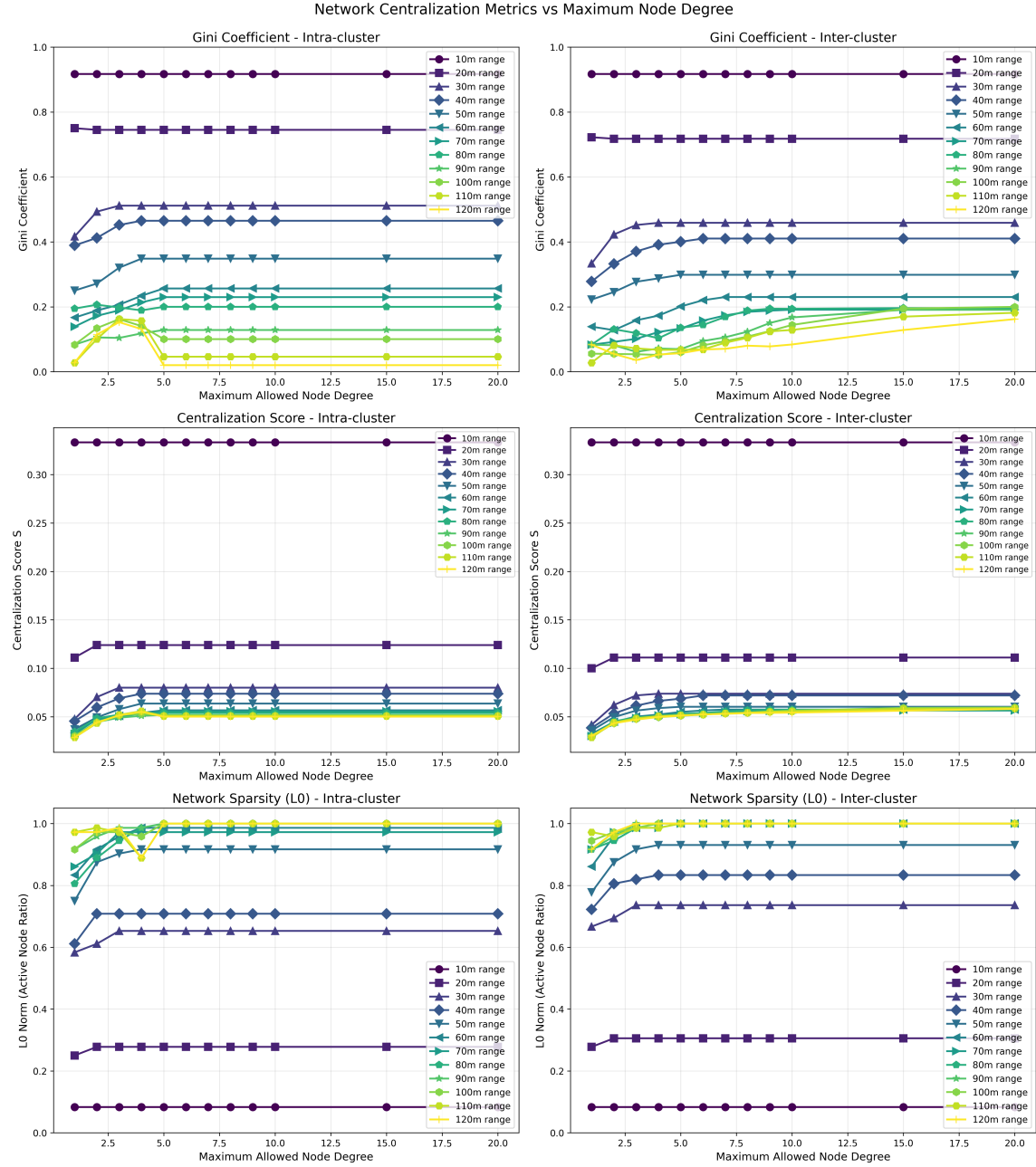


Figure 9: Centralization metrics vs. max degree

Figure 9 shows how the maximum node degree, or the amount of parallel connections allowed per node, affects the different centralization metrics we measured, a subplot per communication mode. In the intra-cluster mode, nodes were only allowed to communicate inside the same cluster, and messages were only created between nodes of the same cluster: there were no connections between nodes of different clusters, and no node attempted to message a node of a different cluster. The inter-cluster mode causes a spike in the number of messages created, because nodes are allowed to communicate both with nodes of the same cluster, as well as outside their cluster: there is, insofar as the cluster membership is considered, no limitation on the communication: a node can communicate with all nodes in its communication radius. The lack of communication limitation enables an exponential rise in the number of messages created: in the intra-cluster mode, 151,904 messages were created, as visible in 5. In the intra-cluster mode, where nodes can establish connections with all nodes within their range upto a maximum configured limit, the message generator generates a message for every pair in every possible permutation of the host list. The same figure shows, that the inter-cluster mode featured over 1.4 million messages. It is important to remark that there is still a limit on the maximum node degree, and each node's links do not churn: once a link has been established in a simulation, it persists until the end of the simulation. At the top we can see the Gini coefficient vs. max node degree, at the middle the Centralization Score vs. max node degree, and at the bottom the L0 norm vs. max node degree, which measured the number of non-zero node degrees.

6 Evaluation

Currently, due to time constraints, some RSSI values are missing from the dataset, and some must be repeated, as evident from figures 2 and 1: the 50 and 100 meter plots are missing, the 80, 90, and 110 meter plots are jittery - some of which due to having an abnormally low received packet count, which indicates the experiment should be repeated in future work. The RSSI values we have provided for the ranges that are complete and stable, are mostly sufficient to derive the necessary path loss parameters for the simulation, as they exhibit the expected power-law distribution and are mostly consistent. Figure 6 interestingly shows that latency increases with distance, which confirms the path-loss model, but exhibits an unexpected result: we expected the 120m radius simulation result to have the highest latency on average, as it created the most messages and suffers the most from the epidemic routing's flooding of messages, but it was dominated by 50-70m ranges, perhaps due to the missing RSSI values at those distances. The centralization metrics plot (Figure 9) summarizes the simulation results, and proves that L0 loss is indeed not enough to characterise centralization score: it has 1.0 for most values. The Gini index and the Centralization score S plots show a positive relation: as the Gini index increases, the S score increases as well, but the curves that dominate the graph are the not the same: the Gini index is dominated by the 40m range, while the S score is dominated by the 120m range.

7 Discussion

Discussion

References

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